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 Studierichting: Informatica
 Oefeningenreeks 5A nr. 7.7

$$\begin{aligned}
 & \int_0^{\frac{\pi}{2}} \frac{dx}{2 + \sin x} \\
 & \Rightarrow \int \frac{dx}{2 + \sin x} = \int \frac{1}{2 + \frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt \\
 & = \int \frac{2}{\left(2 + \frac{2t}{1+t^2}\right)(1+t^2)} dt \\
 & = \int \frac{2}{2(1+t^2) + \frac{2t}{1+t^2}(1+t^2)} dt \\
 & = \int \frac{2}{2 + 2t^2 + 2t} dt \\
 & = \int \frac{1}{t^2 + t + 1} dt \\
 & = \int \frac{1}{\left(t + \frac{1}{2}\right)^2 + \frac{3}{4}} dt \\
 & = \frac{2\sqrt{3}}{3} \operatorname{Bgtan} \frac{2t+1}{\sqrt{3}} + c \\
 & = \frac{2\sqrt{3}}{3} \operatorname{Bgtan} \frac{2 \tan \frac{x}{2} + 1}{\sqrt{3}} + c \\
 & \Rightarrow \int_0^{\frac{\pi}{2}} \frac{dx}{2 + \sin x} = \left[\frac{2\sqrt{3}}{3} \operatorname{Bgtan} \frac{2 \tan \frac{x}{2} + 1}{\sqrt{3}} \right]_0^{\frac{\pi}{2}} \\
 & = \frac{2\sqrt{3}}{3} \left(\frac{\pi}{3} - \frac{\pi}{6} \right) \\
 & = \frac{\pi\sqrt{3}}{9}
 \end{aligned}$$

References